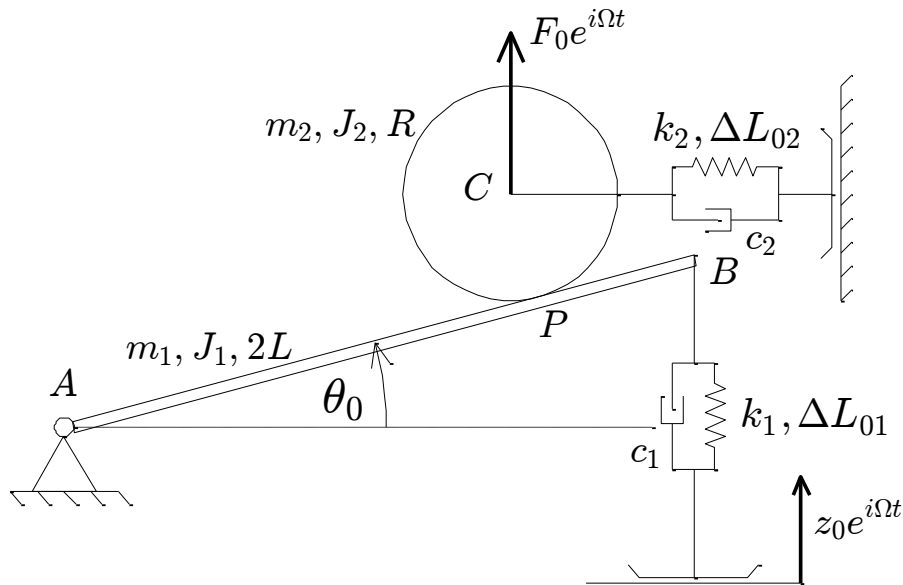


Mechanical Systems Dynamic

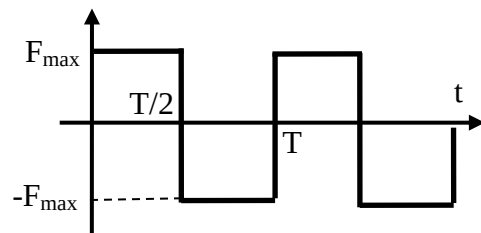
February 02nd, 2017



The mechanical system in the picture lays in the vertical plane and is shown in its equilibrium position. The disk rolls without sliding along the rigid bar AB. The students are requested to:

1. Write the surname, name, identification number and SNxxx code (see note below) on the top of first page of each sheet and leave blank the field named *Nr*. The sheet with uncompleted personal data may be not corrected. For environmental reasons the students are also kindly requested to write all of the four pages of each sheet before using another one.
2. Write the linearized equations of motion around the given static configuration;
3. Compute the natural frequencies and modes of vibrations;
4. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the vertical harmonic force $F_0 e^{i\Omega t}$ at the point C. Output: the rotation of the AB bar and the horizontal component of the displacement of point C. Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your matriculation number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
5. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the vertical constraint displacement $z_0 e^{i\Omega t}$ to the lower extremity of the spring and damper system (k_1, c_1). Output: the vertical displacement of point C and the force transmitted by the spring k_1 . Save the graphical results as SNxxx3.fig and SNxxx4.fig
6. Compute the time history of the vertical displacement of point C for a periodic force applied in C in vertical direction having the waveform shown in the figure below. Save the graphical results as SNxxx5.fig.
7. Considering a harmonic force applied in C in vertical direction, compute the minimum amplitude of the force that causes the loss of contact between the bar and the disk. Write the numerical result on the sheets of the solution, together with the equations used to find this answer.

$m_1 = 10 \text{ kg}$	$k_1 = 5000 \text{ N/m}$	$F_{\max} = 100 \text{ N}$
$m_2 = 5 \text{ kg}$	$k_2 = 7500 \text{ N/m}$	$T = 1.0 \text{ s}$
$J_1 = 0.8 \text{ kgm}^2$	$\Delta L_{01} = -0.0147 \text{ m}$	
$J_2 = 0.15 \text{ kgm}^2$	$\Delta L_{02} = 0.0038 \text{ m}$	
$L = 1.0 \text{ m}$	$c_1 = 3 \text{ Ns/m}$	
$AP = 0.75 \text{ m}$	$c_2 = 5 \text{ Ns/m}$	
$R = 0.25 \text{ m}$	$\theta_0 = \pi/6 \text{ rad}$	



Note: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of the matriculation number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (.fig e .m) have to be compressed (please select either the .zip or the .rar format) in a file named SNxxx.zip (o SNxxx.rar), which has to be uploaded on Beep. Each student is requested to:

1. Connect to the site <https://beep.metid.polimi.it>;
2. Enter the portal using the identification code and password;
3. Select the section Homework of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select the folder *Test Feb. 02 2016- B6.27* or *Test Feb. 02 2016- B6.28* according to the classroom
5. Click the button Add and select Single document;
6. Browse the compressed file SNxxx.zip using the button Browse... (or Sfogliala...)
7. Fill the Title field writing the code SNxxx (e.g. for Bruni Stefano 123456: BS456);
8. Confirm the procedure clicking the button Publish.